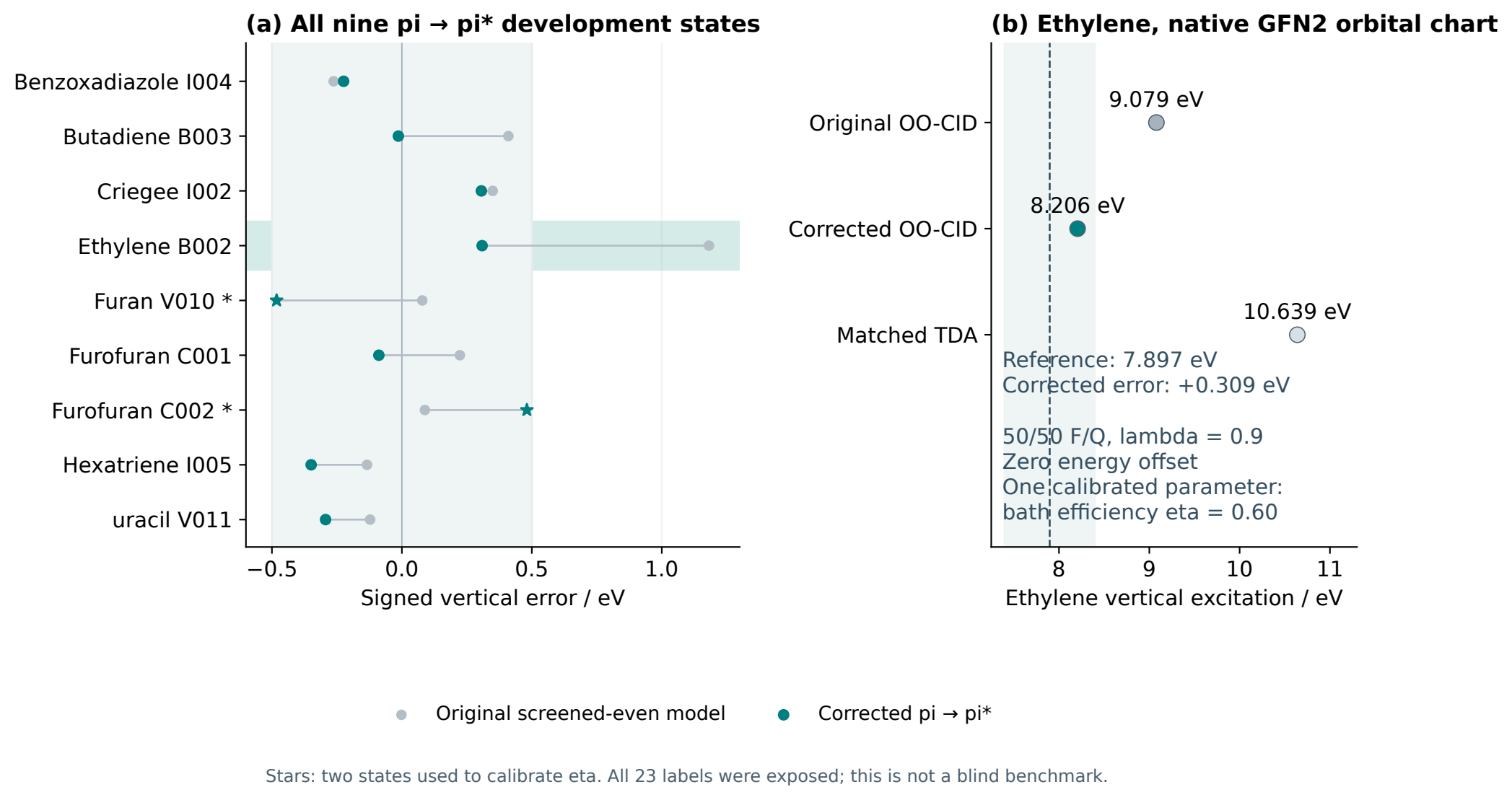
(a) All nine pi → pi* development states
Benzoxadiazole I004
Butadiene B003
Criegee I002
Ethylene B002
Furan V010 *
Furofuran C001
Furofuran C002 *
Hexatriene I005
uracil V011
−0.5
0.0
0.5
1.0
Signed vertical error / eV
(b) Ethylene, native GFN2 orbital chart
Original OO-CID
Corrected OO-CID
Matched TDA
9.079 eV
8.206 eV
10.639 eV
Reference: 7.897 eV
Corrected error: +0.309 eV
50/50 F/Q, lambda = 0.9
Zero energy offset
One calibrated parameter:
bath efficiency eta = 0.60
8
9
10
11
Ethylene vertical excitation / eV
Original screened-even model
Corrected pi → pi*
Stars: two states used to calibrate eta. All 23 labels were exposed; this is not a blind benchmark.